Introduction
Water management is the process of planning, developing, distributing, and managing the optimum use of water resources. It involves both water supply and water demand management.

Section 1: Water Sources

- Surface water (rivers, lakes, reservoirs)
- Groundwater (wells, aquifers)
- Rainwater harvesting

Section 2: Water Distribution
Water is distributed through pipelines, canals, and pumping systems to reach households, industries, and agriculture. Efficient distribution reduces water loss.

Section 3: Water Quality Monitoring
Monitoring water quality ensures safety and compliance with health standards. Key parameters include:

- pH
- Turbidity
- Contaminant levels (heavy metals, chemicals)

Section 4: Leak Detection
Detecting leaks in pipelines is critical to minimize water loss. Techniques include:

- Pressure monitoring
- Acoustic sensors
- Smart meters

Conclusion
Effective water management ensures sustainability and meets the growing demands of urban and rural populations. Combining monitoring, distribution, and leak detection improves efficiency.